11 In May 2019, the distance between Earth and Mars was 3.9×10^7 km.

In May 2019, a signal was sent from Earth to Mars.
Assuming that the signal sent from Earth to Mars travelled at a speed of 3×10^5 km per second,

(a) how long did the signal take to get to Mars?

Time = distance ÷ speed

= (3.9 × 10⁷) ÷ (3 × 10⁵) = 130 s

1 mark
Use distance ÷ speed.

1 mark
130 seconds.

130 seconds

(2)

The speed of the signal sent from Earth to Mars in May 2019 was actually less than 3×10^5 km per second.

(b) How will this affect your answer to part (a)?

At lower speed the journey takes longer.

1 mark
Actual time greater than 130 s.

(1)

(Total for Question 11 is 3 marks)

12 Patrick has to work out the exact value of $64^{\frac{1}{4}}$

Patrick says,

"$\frac{1}{4}$ of 64 is 16 so $64^{\frac{1}{4}} = 16$"

Explain what is wrong with what Patrick says.

(Total for Question 12 is 1 mark)

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